

Alex Siegel

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SUMMARY

Scientist with 15 years of experience in protein biochemistry, structural biology, and biophysics. Expert in characterizing protein-protein interactions and conformational changes by NMR, cryo-EM, FRET, and mass spectrometry with broad skills in other biophysical techniques. Demonstrated ability to lead interdisciplinary projects, mentor students, solve challenging problems, and publish first author papers in high impact journals. Passionate about unraveling molecular mechanisms of proteins to develop novel therapeutic strategies targeting human diseases, particularly neurodegenerative diseases related to chaperone dysfunction and protein misfolding.

RESEARCH EXPERIENCE

* Equal contribution first author publications. Full publication list [here](#).

Postdoctoral Scholar in Chemistry

California Institute of Technology | 2017 - present

- Mapped transient protein-protein interactions between a chaperone and its protein clients by NMR (Siegel*, McAvoy*, et al, *J Mol Biol*, 2020), EPR (Siegel*, Singh*, et al, *RSC*, 2023), and XL-MS (McAvoy, Siegel, et al, *J Biol Chem*, 2018), enabling patent to bioengineer chaperones against disease-causing amyloids (Prov. Patent CIT-9049-P).
- Discovered new clients of a chaperone critical to photosynthesis (Wang et al, *PNAS*, 2018), identified its role in heat protection (Ji*, Siegel* et al, *Nature Plants*, 2021), and uncovered a unique thermostat-like activation mechanism (Siegel et al, *Science Advances*, 2025), informing bioengineering efforts to grow more heat-tolerant crops.
- Solved cryo-EM structure of a human chaperone revealing a unique cage-like assembly, providing a potential drug target for mitochondrial diseases (Gupta*, Lentzsch*, Siegel*, et al, *Science Advances*, 2023).

Graduate Researcher in Biophysics

University of California, Berkeley | 2009 - 2016

- Revealed how stress-responsive bacterial transcription factors force open DNA with NMR (Siegel et al, *J Mol Biol*, 2016) and molecular tweezers (Siegel, *Dissertation*, 2016), providing a potential new antibiotic target.
- Conducted high-pressure NMR studies on deep-sea bacterial transcription factors, revealing mechanistic adaptations.

TECHNICAL SKILLS

Biophysics & Structural Biology. • NMR (15 yrs) • FRET / anisotropy (5 yrs) • Mass spectrometry / XL-MS (5 yrs) • Cryo-EM (5 yrs) • Circular dichroism • Mass photometry • X-ray crystallography • EPR • Molecular dynamics • Molecular tweezers • ITC • MST • SAXS

Molecular Biology & Biochemistry. • Protein expression & purification (15 yrs) • Cloning (Gibson Assembly, QuikChange) • *in vitro* translation • Amber suppression • Western blotting / immunoprecipitation • Bioconjugation • Chemical crosslinking • HPLC • SEC-MALS • Light scattering • High-throughput screening by microplate

Data analysis & software. • Python • R • Matlab • Graphpad • NMR (*MestreNova*, *Topspin*, *NMRPipe*, *NMRView*, *CARA*) • Cryo-EM (*CryoSPARC*, *Relion*) • Mass spectrometry (*ProteinProspector*, *MeroX*) • Structure determination (*PHENIX*, *Coot*) • Structure visualization (*ChimeraX*, *Pymol*) • Structure prediction (*Gromacs*, *AlphaFold*)

LEADERSHIP & COMMUNICATION

- Mentored 20 undergraduate, graduate, and postdoctoral researchers and led student projects to co-authorships.
- Initiated and managed 5 interdisciplinary collaborations; drafted 8 manuscripts, 6 as a first author; contributed to 3 large federally funded grants; presented work at major conferences (EMBO 2021, ASBMB 2022, WPC 2024).
- Vice Chair of Caltech Postdoctoral Association (2018-2022), coordinated postdocs across divisions, organized dozens of career and outreach events, improved onboarding of new postdocs, secured increased career development funds.
- Delegate, UC Berkeley Graduate Assembly (2011-2012), represented biophysics in graduate student policy initiatives.

EDUCATION

Ph.D., Biophysics (Chemical Biology specialization)

University of California, Berkeley | 2016

B.S., Biology & Chemistry (with Honors)

California Institute of Technology | 2008